

Glottal Stop in Forest Nenets: Optimality Theoretic Approach

Background In Forest Nenets (FN; < Samoyedic < Uralic) there is a glottal stop phoneme /ʔ/ present. It, according to Salminen (2007), has two allophones: glottal stop [ʔ], which is an elsewhere allophone (1) and aspiration [h], that occurs before obstruents (2). The data comes from the author’s fieldwork.

- (1) *niʔni* [niʔni] young_man ‘young man’
l'empáʔl'u [l'empiaʔliu] butterfly ‘butterfly’
maʔ [] chum ‘chum’
(2) *mānæ^h-s^o* [] watch-IMP ‘watch!’
ma^h-t^o [] chum-DAT ‘(towards) a chum’
wed'a^hku [] dog ‘dog’

However, there are some cases when an Underlying Representation does not entirely match with the output. Sometimes there is a vowel epenthesis after a glottal stop, that appears when the next after a glottal stop /ʔ/ consonant is either word-final or has a schwa after it, that is not in the Surface Representation. In the example (3) I provide two words, *waʔaw^o* ‘bed’, in which an epenthesis occurs, and *waʔwǎj^o* ‘my bed’, in which it is blocked, as a vowel appears after the glide in the cluster.

- (3) /waʔw^o/ → *waʔaw^o* [waʔaw] bed ‘bed’
/waʔw^oj^o/ → *waʔwǎj^o* [waʔwǎj] bed-POSS.1SG ‘my bed’

Analysis In the present paper I provide Optimality Theoretic (Prince & Smolensky 1993) analysis of the FN data.

I suggest that FN has a constraint that prohibits schwa /^o/ from surfacing in unlicensed positions— ***Schwa** (on schwa surfacing see Salminen (2007)). As there are two allophones, which differ by the [continuant] feature ([ʔ] — [-cont] and [h] — [+cont]), respectively), there is an interaction between the two constraints, ***ʔ[-son]** and **Ident([cont])**. As there are no such output as **maʔt^o*, it might be proposed that ***ʔ[-son]** dominates **Ident([cont])** (see Table 1). There are also **Max-C** and **Dep-V** constraints that ban consonant deletion and vowel epenthesis in unlicensed contexts.

/wéʔk ^o /	*Schwa	Max-C	Dep-V	*ʔ[-son]	Ident([cont])
[wéʔk ^o]	*!				
[wéʔk]				*!	
[wék]		*!			
☞ [wé ^h k]					*
[wéʔek]			*!		

Table 1. Derivation of *wé^hk^o* [wé^hk] neck ‘neck’ from /wéʔk^o/.

For the elsewhere allophone, we can see that no constraints on [ʔ] are violated, which results in an output with [ʔ] (Table 2).

/niʔni/	*Schwa	Max-C	Dep-V	*ʔ[-son]	Ident([cont])
☞ [niʔni]					
[niʔini]			*!		
[ni ^h ni]					*!
[nini]		*!			

Table 2. Derivation of *niʔni* [niʔni] young_man 'young man' from /niʔni/

I propose that there is a constraint connected with the realization of the glottal stop that dominates Dep-V: ***ʔ[+son -voc]{C, #}**. It bans sequences of [ʔ] and a sonorant consonant when no surface vowel is present after a sonorant, resulting in possibility for a vowel epenthesis (Table 3). A constraint against aspiration before sonorants, ***^h[+son]**, also should be proposed, as Dep-V » *ʔ[-son] (see Table 2). ***^h[+son]** dominates *ʔ[-son], so it also dominates Dep-V (see Table 3).

/waʔw ^o /	*Schwa	Max-C	*ʔ[+son]{C, #}	* ^h [+son]	Dep-V	Ident([cont])
[waʔw ^o]	*!					
☞ [waʔaw]					*	
[waʔw]			*!			
[waw]		*!				
[wa ^h w]				*!		*

Table 3. Derivation of *waʔaw^o* [waʔaw] bed 'bed' from /waʔw^o/.